Machine Learning – Lecture Notes

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1 Derivation of Logistic Regression Cost Function

The cost function of the logistic regression is given by:

$$J(\theta) = -\frac{1}{m}\left[\sum_{i=1}^{m} y^{(i)} \log h_\theta(x^{(i)}) + (1-y^{(i)})\log(1-h_\theta(x^{(i)}))\right] + \frac{\lambda}{2m}\sum_{i=1}^{n}\theta_j^2 \tag{1}$$

we know that run the gradient descent, we need to apply to following equation:

$$\theta_j := \theta_j - \alpha\frac{\partial}{\partial\theta}J(\Theta) \tag{2}$$

where we need to plug in the $J(\Theta)$. For this derivation lets drop the regularser term and see what happens.

$$h_\theta(x^{(i)}) = \frac{1}{1+e^{-\theta^T x}} \tag{3}$$

$$g(z) = \frac{1}{1+e^{-z}} \tag{4}$$

The derivative of $g(z)$ is given by:

$$g'(z) = \frac{e^{-z}}{(1+e^{-z})^2} \tag{5}$$

$$= \frac{1+e^{-z}-1}{(1+e^{-z})^2} \tag{6}$$

$$= \frac{1}{(1+e^{-z})}\left(1-\frac{1}{(1+e^{-z})}\right) \tag{7}$$

$$g'(z) = g(z)(1-g(z)) \tag{8}$$

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Now

$$\begin{aligned}
\frac{\partial}{\partial\theta_j}J(\Theta) &= -\frac{1}{m}\frac{\partial}{\partial\theta_j}\left[\sum_{i=1}^{m} y^{(i)}\log h_\theta(x^{(i)}) + (1-y^{(i)})\log(1-h_\theta(x^{(i)}))\right] &(9)\\
&= -\frac{1}{m}\left[\sum_i y^{(i)}\frac{1}{h_\theta(x^{(i)})}\frac{\partial}{\partial\theta_j}h_\theta(x^{(i)}) + (1-y^{(i)})\frac{1}{1-h_\theta(x^{(i)})}\left(-\frac{\partial}{\partial\theta_j}h_\theta(x^{(i)})\right)\right] &(10)\\
&= -\frac{1}{m}\left[\sum_i (A+B)\right] &(11)
\end{aligned}$$

Now using Eqn. 8 in Eqn. 10 for the derivative here we get:

$$\begin{aligned}
A &= y^{(i)}\frac{1}{h_\theta(x^{(i)})}h_\theta(x^{(i)})\left(1-h_\theta(x^{(i)})\right)\times x_j^{(i)} &(12)\\
B &= -(1-y^{(i)})\frac{1}{1-h_\theta(x^{(i)})}\left[h_\theta(x^{(i)})(1-h_\theta(x^{(i)}))\right]\times x_j^{(i)} &(13)
\end{aligned}$$

Now simplifying and putting **A** and **B** back in Eqn. 10, we get:

$$\begin{aligned}
\frac{\partial}{\partial\theta_j}J(\Theta) &= -\frac{1}{m}\left[\sum_i y^{(i)}x_j^{(i)}(1-h_\theta(x^{(i)})) - (1-y^{(i)})x_j^{(i)}h_\theta(x^{(i)})\right] &(14)\\
&= \frac{1}{m}\sum_i\left[y^{(i)}x_j^{(i)} - x_j^{(i)}h_\theta(x^{(i)})\right] &(15)\\
&= -\frac{1}{m}\sum_i\left(y^{(i)}-h_\theta(x^{(i)})\right)x_j^{(i)} &(16)\\
\frac{\partial}{\partial\theta_j}J(\Theta) &= \frac{1}{m}\sum_i\left(h_\theta(x^{(i)})-y^{(i)}\right)x_j^{(i)} &(17)
\end{aligned}$$